

SUMMER EXAMINATION 2026
Machine Learning - Version A
BSIT - 3rd Semester Time: 90 Minutes Max Marks: 80

Name: _____ Roll No.: _____ Date: _____

1. Cross-validation repeatedly splits data into training and
(A) validation folds (B) test labels only
(C) features and targets permanently (D) random classes
2. An RNN processes inputs while maintaining a
(A) hidden state (B) decision stump
(C) centroid (D) confusion table
3. Naive Bayes assumes features are conditionally independent given the
(A) class (B) distance
(C) tree depth (D) reward
4. Stratified splitting preserves
(A) class proportions (B) feature means exactly
(C) time order always (D) duplicate rows
5. A kernel can enable SVMs to model
(A) nonlinear boundaries (B) missing values
(C) test leakage (D) class names
6. A hyperparameter is set
(A) before or outside parameter fitting (B) only by backpropagation
(C) by the test label (D) after deployment only
7. Missing numeric values may be imputed with the
(A) median (B) class name only
(C) file path (D) gradient sign
8. MAE averages the absolute
(A) prediction errors (B) squared features
(C) class labels (D) gradient steps
9. A perceptron is a
(A) single linear threshold unit (B) clustering algorithm
(C) tree ensemble (D) language model
10. PCA finds orthogonal directions ordered by
(A) explained variance (B) class frequency
(C) reward (D) accuracy only
11. Linear regression models a target as a
(A) weighted linear combination (B) tree depth
(C) cluster assignment (D) probability table only
12. Explainability helps stakeholders understand
(A) model predictions (B) network cables
(C) database backups (D) file compression
13. ReLU outputs max of zero and
(A) its input (B) its square
(C) its class (D) its probability
14. DBSCAN identifies dense regions using neighborhood radius and
(A) minimum points (B) tree depth
(C) class weights (D) learning rate
15. Precision equals TP divided by
(A) TP plus FP (B) TP plus FN
(C) TN plus FP (D) all negatives
16. Pooling usually reduces
(A) spatial resolution (B) number of labels
(C) input types (D) training epochs only
17. A decision tree splits data using
(A) feature tests (B) random rewards
(C) vector norms only (D) SQL transactions
18. Grid search evaluates
(A) specified hyperparameter combinations (B) all possible datasets
(C) only one model (D) random gradients
19. The residual in regression is actual value minus
(A) predicted value (B) feature mean
(C) class prior (D) tree depth
20. Supervised learning requires
(A) labeled examples (B) only unlabeled data
(C) no target variable (D) random rewards
21. A reproducible pipeline fixes seeds and records
(A) data, code, and configuration (B) only accuracy
(C) only screenshots (D) only labels
22. L1 regularization encourages
(A) sparse coefficients (B) large weights
(C) more clusters (D) higher variance
23. The learning rate controls
(A) step size in optimization (B) number of classes
(C) test proportion only (D) feature count
24. A discount factor near zero emphasizes
(A) immediate rewards (B) distant rewards
(C) feature scaling (D) test accuracy
25. Logistic regression commonly outputs
(A) a class probability (B) a cluster centroid
(C) a parse tree (D) a continuous image only
26. The sigmoid function maps real values to
(A) between 0 and 1 (B) between -1 and 1 only
(C) all integers (D) unit vectors only
27. Recall equals TP divided by
(A) TP plus FN (B) TP plus FP
(C) TN plus FN (D) all predictions
28. RMSE is the square root of
(A) MSE (B) MAE
(C) R squared (D) variance only
29. Boosting builds an ensemble
(A) sequentially focusing on errors (B) in parallel with no interaction
(C) from one tree (D) only from test data
30. A CNN convolution filter detects
(A) local spatial patterns (B) only class priors
(C) database keys (D) sequence rewards
31. Random forests add randomness by selecting subsets of
(A) features at splits (B) test labels
(C) reward values (D) output classes only
32. A test set should be held out for
(A) final generalization evaluation (B) feature creation
(C) repeated tuning (D) data collection only
33. In Q-learning, $Q(s,a)$ estimates expected return for
(A) an action in a state (B) a feature value
(C) a class label (D) a cluster
34. Unsupervised learning searches for
(A) structure without labels (B) known class names
(C) payment status (D) manual rules
35. Mean squared error averages
(A) squared residuals (B) absolute labels
(C) class probabilities only (D) feature counts
36. The K in K-means is the number of
(A) clusters (B) features
(C) iterations only (D) classes in the test set
37. A larger K in KNN usually makes the decision boundary
(A) smoother (B) more jagged
(C) undefined (D) linear always
38. The first principal component has the greatest
(A) variance (B) mean label
(C) test error (D) number of samples
39. Bayes theorem relates posterior, likelihood, prior, and
(A) evidence (B) variance only
(C) gradient (D) residual
40. A feature is an input variable used by a
(A) model (B) compiler
(C) database index only (D) test runner

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1. A training set is used to
(A) fit model parameters (B) choose final policy only
(C) report unbiased test error (D) encrypt records
2. Backpropagation computes gradients using the
(A) chain rule (B) Bayes rule only
(C) Markov rule (D) Pythagorean theorem
3. A 2 by 2 confusion matrix contains
(A) four outcome cells (B) two models
(C) two features (D) four classes necessarily
4. An outlier is an observation that is
(A) unusually distant from typical values (B) always incorrect
(C) a missing label (D) a duplicate key only
5. Underfitting often indicates a model is
(A) too simple (B) too complex
(C) leaking data (D) using too many labels
6. KNN predicts using the labels of nearby
(A) training points (B) decision nodes
(C) principal components only (D) random samples
7. An epsilon-greedy policy explores randomly with probability
(A) epsilon (B) one minus epsilon always
(C) the learning rate only (D) the discount factor only
8. An activation function adds
(A) nonlinearity (B) database indexing
(C) data leakage (D) a test split
9. Standardization commonly gives a feature mean 0 and
(A) standard deviation 1 (B) range 1
(C) median 1 (D) variance 0
10. R squared measures variance explained relative to a
(A) baseline model (B) confusion matrix
(C) cluster center (D) learning rate
11. Machine learning learns patterns from data to make
(A) predictions or decisions (B) only web pages
(C) fixed hand-coded outputs (D) network cables
12. Overfitting is associated with low training error and high
(A) validation error (B) training accuracy only
(C) bias only (D) regularization strength
13. If a classifier gets 90 correct out of 100, accuracy is
(A) 0.90 (B) 0.10
(C) 0.50 (D) 90.0
14. Hierarchical clustering produces a
(A) dendrogram (B) confusion matrix
(C) ROC curve (D) learning curve only
15. A lower validation loss generally indicates
(A) better validation fit (B) more leakage necessarily
(C) higher bias always (D) fewer samples only
16. An epoch is one pass through the
(A) training data (B) test labels only
(C) hyperparameter grid (D) confusion matrix
17. Entropy measures
(A) impurity or uncertainty (B) distance to origin
(C) learning rate (D) number of layers
18. Gradient boosting fits new learners to
(A) current residual errors (B) feature names
(C) random labels (D) test predictions only
19. For TP=12 and FN=3, recall is
(A) 0.80 (B) 0.20
(C) 0.75 (D) 0.60
20. Accuracy can be misleading when classes are
(A) imbalanced (B) balanced
(C) continuous (D) standardized
21. Specificity is the true negative rate, TN divided by
(A) TN plus FP (B) TN plus FN
(C) TP plus FP (D) all positives
22. One-hot encoding represents a category with
(A) binary indicator columns (B) a single ordered rank
(C) a text paragraph (D) a distance matrix only
23. The ROC AUC summarizes ranking quality across
(A) classification thresholds (B) learning rates only
(C) cluster counts (D) feature encodings
24. Model drift means the data or relationship changes after
(A) deployment (B) normalization
(C) training split (D) label encoding
25. LSTM gates regulate information in the
(A) cell state (B) test set
(C) feature scaler (D) tree root
26. Support vectors are training points closest to the
(A) decision boundary (B) centroid
(C) root node (D) mean label
27. Pruning a tree helps reduce
(A) overfitting (B) missing values
(C) class count (D) feature scaling
28. Fairness evaluation checks whether outcomes differ across
(A) relevant groups (B) random file names
(C) feature columns only (D) epochs only
29. Early stopping can reduce overfitting by halting when validation performance
(A) stops improving (B) reaches zero training loss
(C) has more features (D) uses no labels
30. The target in a regression problem is usually
(A) continuous (B) a cluster ID only
(C) a token sequence (D) a Boolean rule only
31. L2 regularization penalizes the
(A) sum of squared weights (B) number of classes
(C) absolute residual only (D) test size
32. Min-max scaling typically maps values to
(A) a chosen interval such as 0 to 1 (B) integer labels
(C) random signs (D) probabilities always
33. Monitoring should track performance, latency, and
(A) input or output drift (B) font size
(C) source comments (D) tree names
34. Feature selection removes
(A) unhelpful input variables (B) all target labels
(C) test cases (D) model predictions
35. Gradient descent moves parameters opposite the
(A) gradient of loss (B) data mean
(C) test set (D) class label
36. An SVM seeks a separating hyperplane with a large
(A) margin (B) entropy
(C) learning rate (D) cluster count
37. Bagging trains models on
(A) bootstrap samples (B) one fixed sample only
(C) ordered labels (D) future data
38. K-means alternates assignment and
(A) centroid update (B) tree pruning
(C) gradient clipping (D) label encoding
39. Reinforcement learning uses actions, states, and
(A) rewards (B) SQL joins
(C) parse trees (D) static labels only
40. Data leakage occurs when information from the future or target
(A) enters training improperly (B) is normalized
(C) is shuffled (D) is missing